

Zen Nguyen

(672) 673-2613 | ttn54@sfu.ca | linkedin.com/in/zennguyen1305 | github.com/ttn54 | zennguyen.me

TECHNICAL SKILLS

- **Languages:** C++, Java, Python, Go
- **Backend & Systems:** RESTful APIs, gRPC, TCP Sockets, Git, Linux (WSL)

TECHNICAL PROJECTS

TitanSwarm (Autonomous AI Co-Pilot) & TitanStore (Raft Database) Jan 2026 - Present
Personal Project · Python, Go, LangChain, FAISS

- Enhanced **complex software systems** by architecting a distributed memory bank in Go using the Raft consensus protocol, ensuring high availability and fault tolerance.
- Improved **functional excellence** by implementing a custom Write-Ahead Log (WAL) that guarantees application state durability across network failures.
- Optimized data throughput by designing a high-throughput TCP client API, enabling concurrent real-time status updates for 100+ users.
- Automated end-to-end workflows by building a zero-hallucination RAG engine, delivering tailored outputs while maintaining strict data integrity.

Gridlock Casino (2D Arcade Engine) Feb 2026 - Present
Collaborative Project · Java, Swing, Maven, JUnit

- Demonstrated **problem-solving acumen** by architecting a custom 2D grid-based engine in Java, managing 60-FPS rendering cycles and complex MVC state transitions.
- Ensured high-quality software delivery by managing dependency resolution with Apache Maven and validating core engine logic through comprehensive JUnit testing.

SFU Course Tracker (sfucourseplanner.me) Nov 2025 - Jan 2026
Personal Project · Python, Docker, AWS, FastAPI

- Modernized infrastructure by migrating a full-stack platform to AWS (EC2) with Docker container orchestration, ensuring **seamless integration and performance** across environments.
- Engineered a robust scraping pipeline using Asyncio and HTTPX, increasing data throughput by 10x while maintaining strict rate-limiting constraints.
- Developed a deterministic Abstract Syntax Tree (AST) parser to tokenize nested boolean prerequisite strings, solving complex data structure challenges.
- Optimized read performance for complex queries by designing a normalized SQLAlchemy schema with JSON-type columns for recursive tree storage.

EDUCATION

Bachelor of Science, Computing Science May 2025 - Present
Simon Fraser University

- CGPA: 3.74 / 4.33
- Relevant Coursework: Discrete Math (A+), Computer Systems(A), Data Structures & Programming (B+), Linear Algebra(A-).
- Langara College | Associate of Science, Computer Science Jan 2024 - Apr 2025

WORK EXPERIENCE

Server · Pho Goodness Restaurant Jan 2024 - Present
Burnaby, BC

- Streamlined high-volume operations by coordinating effectively with diverse teams under pressure, ensuring **attention to detail** for 100+ guests per shift.
- Maintained high performance standards while balancing academic responsibilities, demonstrating a logical approach to managing competing priorities.